

Malvern Madondo

CONTACT INFORMATION

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EDUCATION

Emory University, Atlanta, GA *Aug, 2019 - present*
PhD Student, Computer Science and Informatics
The College of St. Scholastica, Duluth, MN *Aug, 2015 - May, 2019*
B.S., Computer Science/Information Systems and B.A., Mathematics

RESEARCH INTERESTS AND SKILLS

Interests: Reinforcement Learning; Optimal Control; Neuroscience; Fairness
Skills: Python(incl. PyTorch, TensorFlow, NumPy); Java; MATLAB; SQL
Version Control: Git; Mercurial **OS:** Windows; macOS; Linux

RESEARCH EXPERIENCE

Closed-loop neurostimulation via Reinforcement Learning and Optimal Control
Machine Learning and Inverse Problems (MLIP), Emory University *Jun, 2020 - present*

- Thesis research on Reinforcement Learning-based approaches to solving optimal control problems, particularly in systems with nonlinear, unknown, and stochastic dynamics. Advised by [Lars Ruthotto](#).

INTERNSHIP EXPERIENCE

IBM Research - Responsible and Inclusive Technologies Team, Yorktown Heights, NY
PhD Research Intern *May - Aug, 2022*

- Under the IBM Sustainability Accelerator effort (in collaboration with Texas A&M AgriLife team), I implemented a process-based crop growth OpenAI gym environment and developed Reinforcement Learning and classical methods for crop yield optimization .

IBM Research - Responsible and Inclusive Technologies Team, *remote*
PhD Research Intern *May - Aug, 2021*

- Conducted research at the intersection of fairness, representation learning, and reinforcement learning, with the goal of mitigating bias in patients representation in healthcare settings.

Facebook - Crisis Response/Safety Check Team, Menlo Park, CA
Software Engineering Intern *May - Aug, 2018*

- Developed the msite version of Community Help, a platform which allows users to find and offer help during a crisis, to Facebook's mobile site. Also built a Mark Safe QP on FB's web platform.

Facebook - Facebook University for Engineering, Menlo Park, CA
Software Engineering Intern *Jun - Aug, 2017*

- Developed from scratch an iOS app that integrated location-based facts, Tinder-style fact sharing, localization, image recognition (utilizing IBM Watson API) and Wikipedia descriptions.

PUBLICATIONS

Madondo, M., Ruthotto, L., and Au Yong, N. **Towards Closed-loop Neurostimulation via Optimal Control**. To appear in *SIAM MDS (2022)*.
Madondo, M., Shahn, Z., Ghalwash, M., and Meyer, P. **Fair Reinforcement Learning via Fair Representations**. *In preparation (2022)*
Madondo, M. and Gibbons, T. **Learning and Modeling Chaos Using LSTM Recurrent Neural Networks**. *Midwest Instruction and Computing Symposium (MICS 2018)*.
Madondo, M., Gomez, DM., and Ciernia, N. **Hitchhiker's Guide to Computer Science for Social Good** *Midwest Instruction and Computing Symposium (MICS 2018)*.

HONORS AND AWARDS

2021 Google PhD Fellowship recipient in Computational Neural and Cognitive Sciences
2021 IBM Scholar, CMD-IT/ACM Richard Tapia Celebration of Diversity in Computing Conference
2020 Computing Research Association Grad Cohort Workshop for IDEALS scholarship recipient